Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions

2. Dataset Summary

. - Rows: 3,900
- Columns: 18
- Key Features:
- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
- Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3.Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.

- Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Prom Cod Use
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Ye
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Ye
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Ye
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Ye
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Ye

Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
Yes	14	Venmo	Fortnightly
Yes	2	Cash	Fortnightly
Yes	23	Credit Card	Weekly
Yes	49	PayPal	Weekly
Yes	31	PayPal	Annually

● **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

● **Column Standardization:** Renamed columns to snake case for better readability and documentation.

● **Feature Engineering:** ○ Created age_group column by binning customer ages.

○ Created purchase_frequency_days column from purchase data.

● **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

● **Database Integration**: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1.**Subscription Rate:** Calculated the subscription rate using conditional aggregation to identify the proportion of customers who opted for subscription services.

	subscribers bigint	total_customers bigint	subscription_rate numeric
1	1053	3900	27.00

2.Customer Segmentation: Segmented customers into New, Returning, and Loyal categories based on their previous purchase behavior to analyze customer engagement and retention patterns.

	customer_segment text
1	Loyal
2	Returning
3	Loyal
4	Loyal
5	Loyal
6	Loyal
7	Loyal
8	Loyal
9	Loyal

3. **Customer Segment-wise Subscription Analysis**: Compared subscription behavior across customer segments to identify which group (New, Returning, or Loyal) contributes most to subscriptions.

	customer_segment text	subscription_status text	total bigint
1	Loyal	No	2518
2	Returning	Yes	95
3	Loyal	Yes	958
4	Returning	No	329

4.Top Product Categories by Purchase Volume: Identified the most popular product categories by analyzing purchase frequency across all transactions.

	category text	total_purchases bigint
1	Clothing	1737
2	Accessori...	1240
3	Footwear	599
4	Outerwear	324

5.**Impact of Discounts on Subscription Behavior**: Analyzed how discount offers influence customer subscription decisions by comparing subscription patterns with and without discounts.

	discount_applied text	subscription_status text	total bigint
1	Yes	Yes	1053
2	No	No	2223
3	Yes	No	624

6.Top Locations by Total Sales: Identified high-performing regions by analyzing total sales across different locations to understand geographic revenue distribution.

	location text	total_sales numeric
1	Montana	5784
2	Illinois	5617
3	California	5605
4	Idaho	5587
5	Nevada	5514
6	Alabama	5261
7	New York	5257
8	North Dakota	5220
9	West Virginia	5174

7.Customer Segment with Highest Subscription Rate: Evaluated subscription rates across customer segments to identify which group shows the highest conversion to subscription services.

	customer_segment text	subscription_rate numeric
1	Loyal	27.5604142692750288
2	Returning	23.4604105571847507
3	New	18.0722891566265060

5. Dashboard in Power BI

6. Business Recommendations

● **Improve Subscription Conversion:** With only ~27% subscription rate, implement targeted campaigns (emails, offers) to convert non-subscribed customers.

● **Focus on Loyal Customers:** Since loyal customers show higher subscription rates, businesses should prioritize retention strategies and exclusive benefits for them.

● **Leverage High-Performing Categories:** Invest more in top-performing categories (like clothing) through promotions, inventory expansion, and marketing campaigns.

●**Use Discounts Strategically:** Since discounts positively influence subscriptions, offer limited-time deals to increase customer engagement and conversions.

● **Target High-Sales Locations:** Focus marketing and logistics efforts on top-performing locations to maximize revenue and expand market reach.